MILITARY STANDARD

MIL-STD-1386-12
(NAVY)

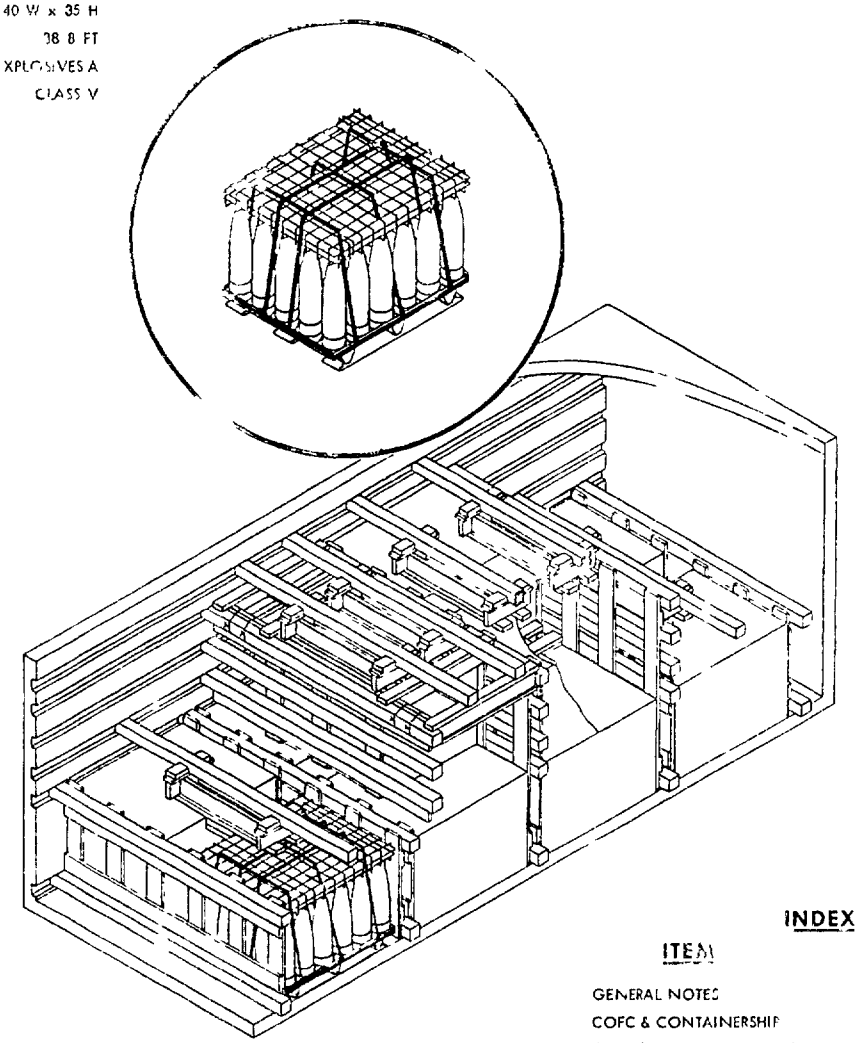
NOVEMBER 1972

LOADING OF HAZARDOUS MATERIALS AND ASSOCIATED ITEMS IN MILVAN CONTAINERS

PROJECTILE, 6 INCH AAC OR HC WITH PROTECTIVE CAP MK 1 & 3 IN PALLET ADAPTER MK 20 FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING	WR-5478
WEIGHT	3440 LBS
DIMENSIONS	48 L x 40 W x 35 H
CUBE	38.8 FT
DOT HAZARD CLASSIFICATION	EXPLOSIVES A
COAST GUARD CLASS	CLASS V



INDEX

ITEM	PAGE
GENERAL NOTES	2
COFC & CONTAINERSHIP	2 & 3
COFC/TOFC CONTAINERSHIP & HIGHWAY	4 & 5
DETAILS	6

NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX OF STANDARDS, MIL-HDBK-236

AUTHORIZED FOR TRAILER-ON-FLATCAR (TOFC), CONTAINER-ON-FLATCAR (COFC), CONTAINERSHIP AND HIGHWAY SHIPMENTS

SIGNATURE — ORDNANCE (SEA) — DATE 10/17/72

SIGNATURE — ORDNANCE, BY DIRECTION — DATE 3/27/73

APPROVED BY US COAST GUARD

DATE 1-31-73

APPROVED BY BUREAU OF EXPLOSIVES

DATE 1/23/73

ORIGINATOR — SIGNATURE

NAVAL [illegible] AND HANDLING LABORATORY
NAVAL AMMUNITION DEPOT CRANE, INDIANA

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:11Z
Check the source to verify that this is the current version before use.

MIL STD 138C (NAVY)

GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR LOADING THE MILVAN CONTAINER WITH FLEET ISSUE UNIT LOAD OF 6 INCH MAC OR HC PROJECTILES (WITH PROTECTIVE CAPS MK 1 & 3) IN PALLET ADAPTER MK 20. THE UNIT LOAD SHALL BE IN ACCORDANCE WITH WR-54/B.
2. THE MILVAN SHALL BE EQUIPPED WITH A MECHANICAL LOAD-BRACING SYSTEM THAT MEETS THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6C AND SHIPPED ON FLATCARS THAT MEET THE REQUIREMENTS OF PAMPHLET 6C.
3. IN ADDITION TO THE BELT RAILS REQUIRED BY PAMPHLET 6C, THE MILVAN CONTAINER HAS BELT RAILS AT THE 72 AND 83 INCH HEIGHTS FROM THE CONTAINER FLOOR.
4. CONTAINERS ARE EQUIPPED WITH 25 CROSSMEMBERS. UNUSED CROSSMEMBERS MUST REMAIN WITH THE CONTAINER AND SHALL BE FASTENED TO BELT RAILS FOR SHIPMENT.

CAUTION

WHEN POSITIONING CROSSMEMBERS, MAKE A VISUAL CHECK TO ENSURE THAT CROSSMEMBERS ARE LOCKED SECURELY IN PLACE.

5. THE DUNNAGE SPECIFIED IS OF NOMINAL SIZE. FOR EXAMPLE, 1 x 4 MATERIAL IS ACTUALLY 3/4 THICK BY 3 5/8 WIDE AND 2 x 6 MATERIAL IS ACTUALLY 1 5/8 THICK BY 5 1/2 WIDE.
6. WHEN THE VOID BETWEEN THE CROSSMEMBERS AND THE SEPARATOR GATE IS GREATER THAN ONE INCH USE FILL MATERIAL TO SHIM BETWEEN CROSSMEMBER AND GATE AS SHOWN IN DETAIL N, PAGE 6.
7. APPLICABLE MATERIAL SPECIFICATIONS
   - DUNNAGE LUMBER - FED SPEC MM-L-751
   - NAILS - FED SPEC FF-N-105, COMMON CEMENT COATED
   - PLYWOOD - COMMERCIAL STANDARD C545 GRADE INTERIOR C-D
8. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS INSIDE THE CONTAINER IN AN ACCESSIBLE AREA, CLOSE AND SEAL CONTAINER DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF THE CONTAINER FOR THE APPROPRIATE MODE OF TRANSPORTATION AS PRESCRIBED IN OP 2165 VOL 1.

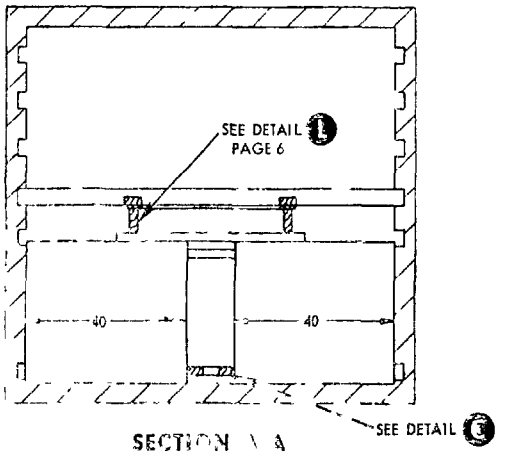
COTC AND CONTAINERSHIP ONLY

1. THE CONTAINERLOAD CONSISTS OF 11 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS DOCUMENT.

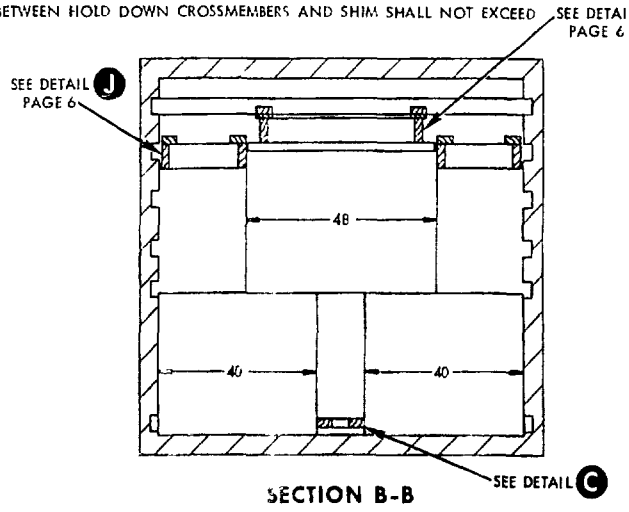
WARNING

BECAUSE THE WEIGHT LOADED INTO THIS CONTAINER EXCEEDS 34,000 LBS, THE MILVAN CHASSIS CAN NOT BE USED TO TRANSPORT THE LOADED CONTAINER ON PUBLIC HIGHWAYS. THE LOADED CONTAINER MAY BE TRANSPORTED OVER DEPOT ROADS USING MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES OR COMMERCIAL EQUIPMENT OF APPROPRIATE CAPACITY.

2. SWAY BRACE, DETAIL C, IS ASSEMBLED IN PLACE. PIECE 5 IS POSITIONED INBOARD OF THE OUTER LEGS OF THE PALLET SKIDS AND PIECE 6 IS POSITIONED AGAINST THE PALLET LEGS. NAIL PIECE 6 TO PIECE 5 WITH THREE 10d NAILS PER JOINT. CUT PIECE 7 TO FIT BETWEEN PIECES 6 AND NAIL TO PIECE 5 WITH THREE 10d NAILS.
3. LENGTH OF CROSSMEMBERS PIECE 9 SHALL BE SUFFICIENT TO CAUSE DETAIL J TO FILL THE VOID BETWEEN THE UNIT LOAD AND THE CONTAINER SIDEWALL. POSITION ASSEMBLY WITH PIECE 11 AGAINST CONTAINER SIDEWALL. TWIST TIE BOTH ENDS OF ASSEMBLY TO CROSSMEMBER WITH 16 GAUGE SOFT ANNEALED WIRE.
4. SHIM, PIECE 15, MAY CONSIST OF MORE THAN ONE PIECE OF DIFFERENT THICKNESSES AND SHOULD CAUSE THE ASSEMBLY TO FILL THE VOID BETWEEN THE LADING AND THE CROSSMEMBER. VOIDS BETWEEN HOLD DOWN CROSSMEMBERS AND SHIM SHALL NOT EXCEED ONE-HALF INCH.

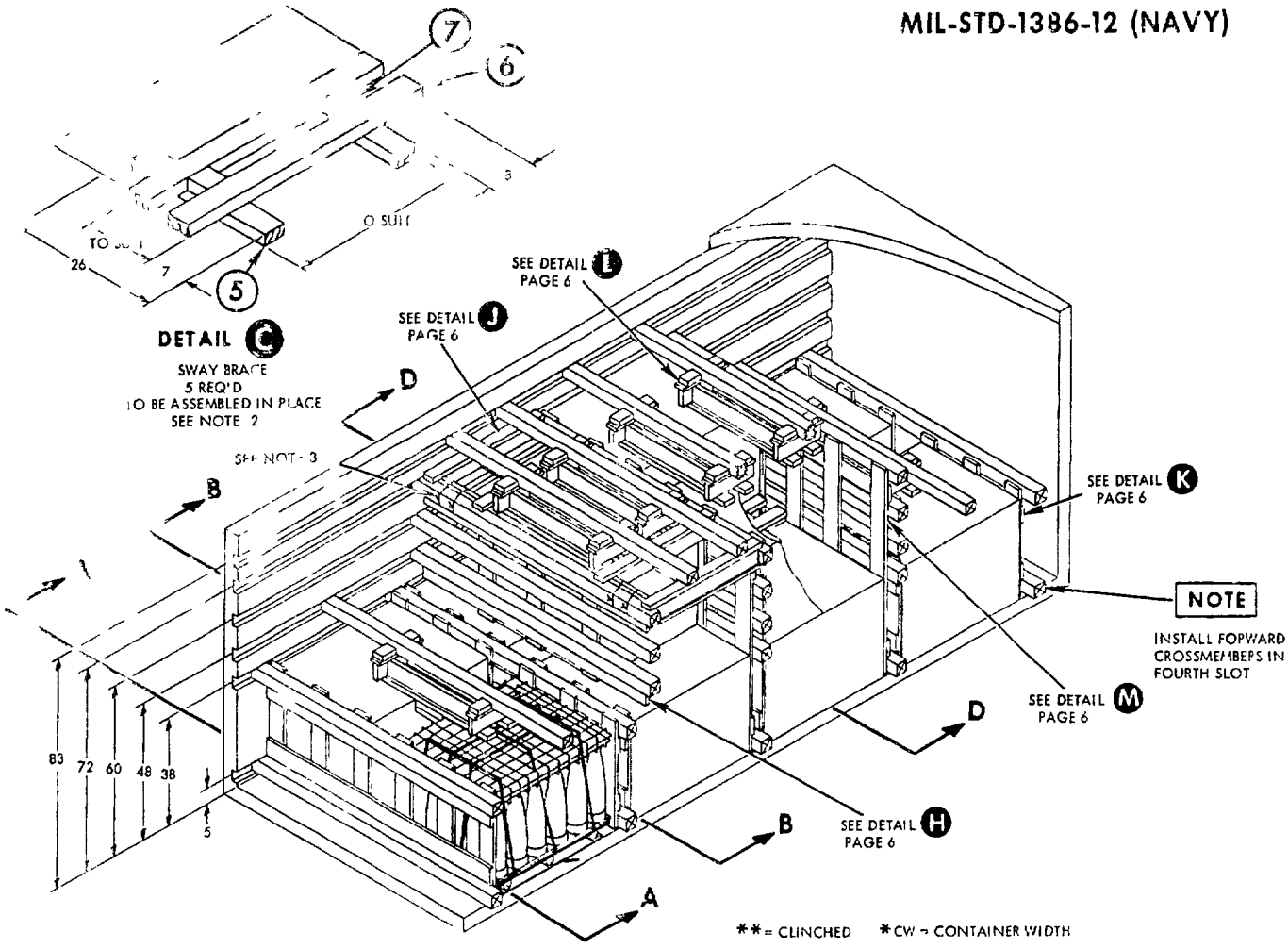


SECTION A-A



SECTION B-B

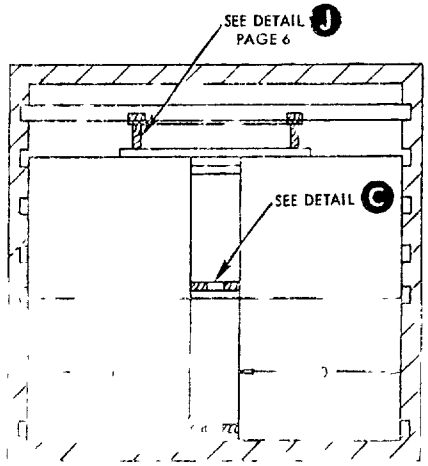
Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:11Z
Check the source to verify that this is the current version before use.



**= CLINCHED *CW = CONTAINER WIDTH

CONTAINER LOAD DATA

NUMBER OF UNIT LOADS	11
NUMBER OF CROSSMEMBEPS REQ'D	25
LOAD WEIGHT	37 840 LBS
DUNNAGE WEIGHT	932 LBS
CONTAINER WEIGHT	5,785 LBS
GROSS WEIGHT	44,557 LBS



PIECE NO.	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
		ISO MATERIALS & NAILING DATA				
19	VERTICAL	1 x 6 x 72	18	18	3/JOINT	6d**
18	HORIZONTAL	1 x 6 x CW-1*	6	SEE 19	-	-
17	SPACER	2 x 4 x TO SUIT	24	12, 17	3	10d
16	RETAINER	2 x 4 x 6	24	15	2	10d
15	SHIM	4 WIDE x THICKNESS TO SUIT x 18	AS REQ'D	13	3	10d
14	SUPPORT	2 x 6 x TO SUIT	12	SEE 12 & 13	-	-
13	RISER	2 x 6 x 18	12	14	2/JOINT	10d
12	LOWER BEARING PIECE	2 x 4 x 47	12	14 13	3 2/JOINT	10d 10d
11	SPACER	1/2 PLYWOOD 1 WIDE x 39 LONG	2	8	4	6d
10	SUPPORT	2 x 4 x 45 1/2	4	9 8	2/JOINT 6	10d 10d
9	CROSSMEMBER	2 x 6 x TO SUIT	6	SEE 8	SEE NOTE 3	-
8	LONGITUDINAL	2 x 6 x 39	4	9	2/JOINT	10d
7	SPACER	2 x 4 x TO SUIT	10**	5	3	10d
6	STRINGER	2 x 4 x 47	10**	5	3/JOINT	10d
5	SUPPORT	2 x 4 x 26	10**	SEE 6 & 7	-	-
4	VERTICAL	1 x 6 x 36	3	3	3/JOINT	6d**
3	HORIZONTAL	1 x 6 x CW-1*	2	SEE 4	-	
2	VERTICAL	1 x 6 x 38	30	1	3/JOINT	6d**
1	HORIZONTAL	1 x 6 x CW-1*	10	SEE 2	-	-

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:11Z
Check the source to verify that this is the current version before use.

TOFC/COFC, CONTAINERSHIP & HIGHWAY

1 THE CONTAINERLOAD CONSISTS OF 9 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS DOCUMENT PROVIDED THE GROSS WEIGHT LOADED INTO THIS CONTAINER DOES NOT EXCEED 34,000 LBS THE 20 FT MILVAN CHASSIS WITH DOUBLE AXLES MAY BE USED TO TRANSPORT THE LOADED CONTAINER VIA TOFC AND HIGHWAY COMMERCIAL EQUIPMENT OF APPROPRIATE CAPACITY MAY ALSO BE USED FOR HIGHWAY MOVEMENT

NOTE

IN CERTAIN STATES THE GROSS VEHICLE WEIGHT (G V W) (CONTAINER CHASSIS AND TRACTOR) IS LIMITED BY THE DISTANCE BETWEEN THE 1ST AND THE LAST AXLES OF THE ASSEMBLED RIG CARE SHOULD BE TAKEN TO SEE THAT THE PROPOSED G V W DOES NOT EXCEED THESE LIMITATIONS IF IT DOES, THE WEIGHT MAY BE REDUCED BY REDUCING THE NUMBER OF UNIT LOADS IN THE CONTAINER AS FOLLOWS

A OMIT THE UNIT LOAD IN THE 2ND LAYER

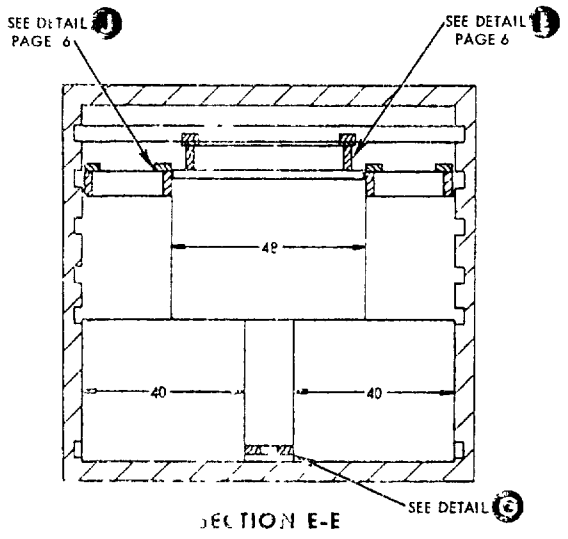
B OMIT ADDITIONAL UNIT LOADS STARTING AT THE REAR OF THE CONTAINER

WHEN REDUCING THE NUMBER OF UNIT LOADS IN A CONTAINER, THE REMAINING UNIT LOADS SHALL BE BLOCKED AND BRACED USING THE PRINCIPLES OF THIS DOCUMENT

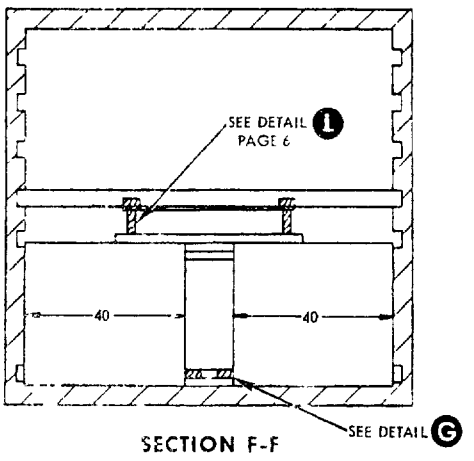
2 SWAY BRACE, DETAIL G IS ASSEMBLED IN PLACE PIECE 5 IS POSITIONED INBOARD OF THE OUTER LEGS OF THE PALLET SKIDS AND PIECE 6 IS POSITIONED AGAINST THE PALLET LEGS NAIL PIECE 6 TO PIECE 5 WITH THREE 10d NAILS PER JOINT CUT PIECE 7 TO FIT BETWEEN PIECES 6 AND NAIL TO PIECE 5 WITH THREE 10d NAILS

3 LENGTH OF CROSSMEMBERS PIECE 9 SHALL BE SUFFICIENT TO CAUSE DETAIL J TO FILL THE VOID BETWEEN THE UNIT LOAD AND THE CONTAINER SIDEWALL POSITION ASSEMBLY WITH PIECE 11 AGAINST CONTAINER SIDEWALL TWIST TIE BOTH ENDS OF ASSEMBLY TO CROSSMEMBER WITH 16 GAUGE SOFT ANNEALED WIRE

4 SHIM, PIECE 15, MAY CONSIST OF MORE THAN ONE PIECE OF DIFFERENT THICKNESSES AND SHOULD CAUSE THE ASSEMBLY TO FILL THE VOID BETWEEN THE LADING AND THE CROSSMEMBER VOIDS BETWEEN HOLD DOWN CROSSMEMBERS AND SHIM SHALL NOT EXCEED ONE HALF-INCH

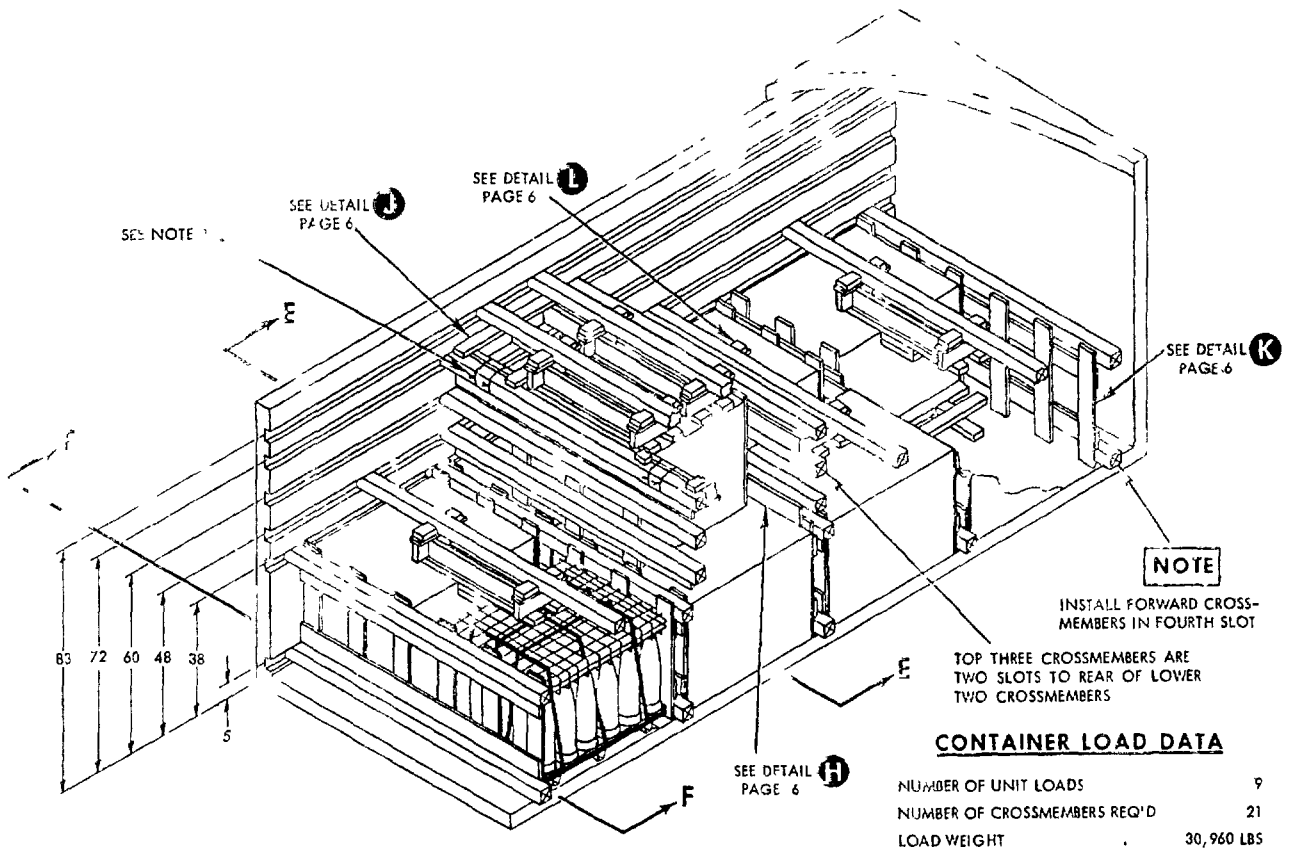


SECTION E-E



SECTION F-F

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:11Z
Check the source to verify that this is the current version before use.



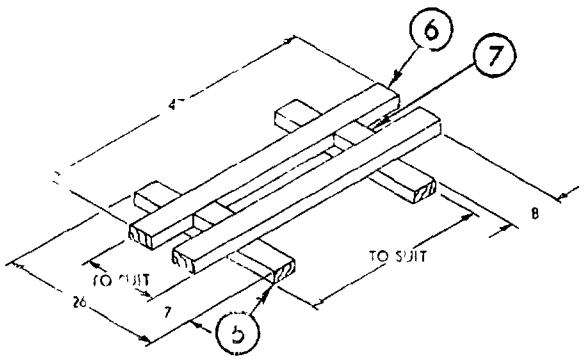
CONTAINER LOAD DATA

NUMBER OF UNIT LOADS	9
NUMBER OF CROSSMEMBERS REQ'D	21
LOAD WEIGHT	30,960 LBS
DUNNAGE WEIGHT	820 LBS
CONTAINER WEIGHT	5 785 LBS
GROSS WEIGHT	37,565 LBS

*CW - CONTAINER WIDTH **CLINCHED

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
17	SPACER	2 x 4 x TO SUIT	20	12, 17	3	10d
16	RETAINER	2 x 4 x 6	20	15	2	10d
15	SHIM	4 WIDE x THICKNESS TO SUIT x 18	AS REQ'D	13	3	10d
14	SUPPORT	2 x 6 x TO SUIT	10	SEE 12 & 13	-	-
13	RISER	2 x 6 x 18	10	14	2/JOINT	10d
12	LOWER BEARING PIECE	2 x 4 x 47	10	14 13	3 2/JOINT	10d 10d
11	SPACER	1/2 PLYWOOD 1 WIDE x 39 LONG	2	8	4	6d
10	SUPPORT	2 x 4 x 45 1/2	-	9 8	2/JOINT 4	10d 10d
9	CROSSMEMBER	2 x 6 x TO SUIT	6	SEE 8	SEE NOTE 3	-
8	LONGITUDINAL	2 x 6 x 39	4	9	2/JOINT	10d
7	SPACER	2 x 4 x TO SUIT	8	5	3	10d
6	STRINGER	2 x 4 x 47	8	5	3/JOINT	10d
5	SUPPORT	2 x 4 x 26	8	SEE 6 & 7	-	-
4	VERTICAL	1 x 6 x 36	6	3/JOINT	3/JOINT	6d**
3	HORIZONTAL	1 x 6 x CW*-1	4	SEE 4	-	-
2	VERTICAL	1 x 6 x 38	48	1	3/JOINT	6d**
1	HORIZONTAL	1 x 6 x CW*-1	16	SEE 2	-	-

LIST OF MATERIALS & NAILING DATA



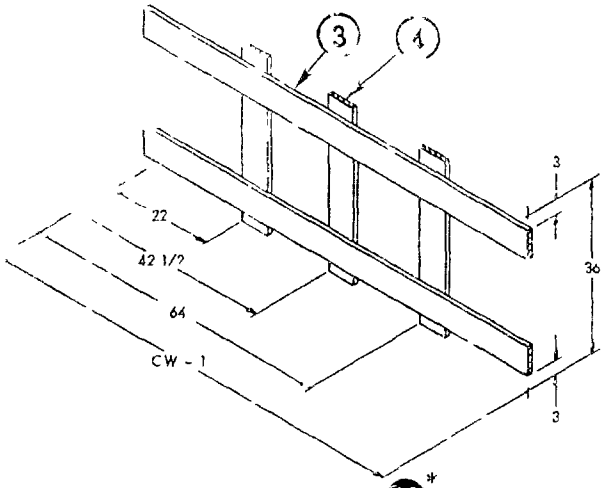
DETAIL G

SWAY BRACE
4 REQ'D
NOTE: ASSEMBLED W PLAY
SEE PAGE 6

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:11Z
Check the source to verify that this is the current version before use.

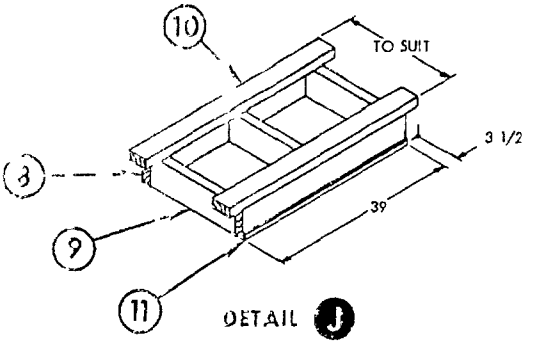
MIL [illegible]

DETAILS



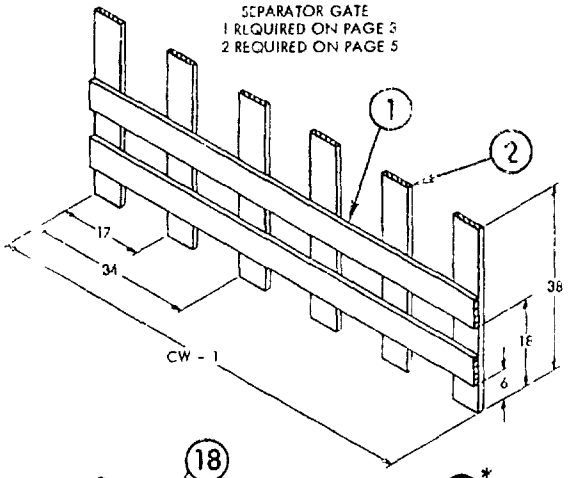
DETAIL H*

SEPARATOR GATE
1 REQUIRED ON PAGE 3
2 REQUIRED ON PAGE 5



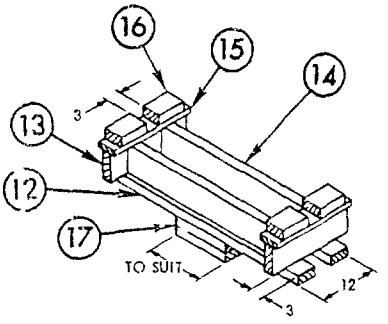
DETAIL J

SWAY BRACE
2 REQUIRED FOR PAGES 3 & 5
SEE NOTE 2, PAGE 2 OR 4



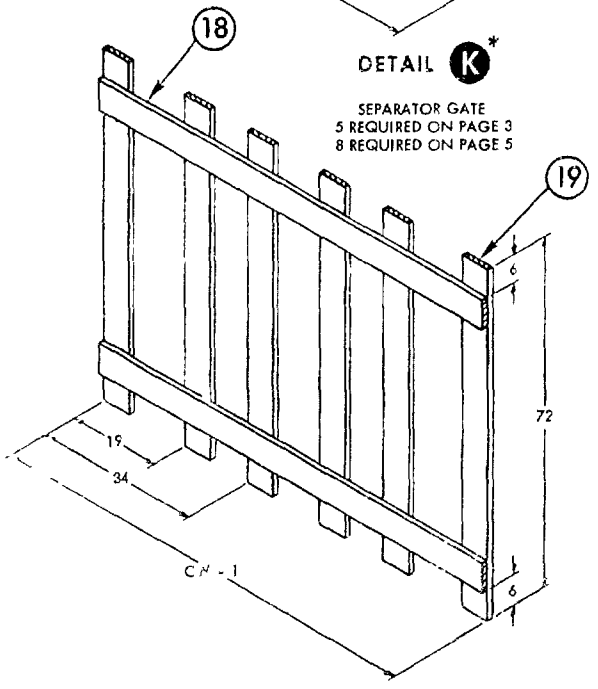
DETAIL K*

SEPARATOR GATE
5 REQUIRED ON PAGE 3
8 REQUIRED ON PAGE 5



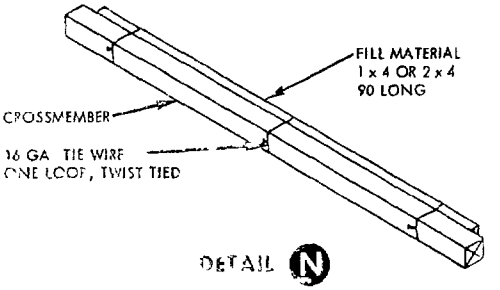
DETAIL L

HOLDDOWN SHIM ASSEMBLY
6 REQUIRED ON PAGE 3
5 REQUIRED ON PAGE 5
SEE NOTE 4, PAGE 2 OR 4



DETAIL M*

SEPARATOR GATE
3 REQUIRED ON PAGE 3



DETAIL N

FILL MATERIAL
SEE GENERAL NOTE 6, PAGE 2

* POSITION SEPARATOR GATE WITH VERTICALS AGAINST UNIT LOAD

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T00:11Z
Check the source to verify that this is the current version before use.